CAO-DF - Sequence Diagram: randomized blocks of measured runs (A1 / A2 / E6)

Experiment layer

run_a1 / run_a2 / run_validation

Infrastructure

spark_config get_spark_session()

instrumentation run_and_measure()

_ResourceMonitor (psutil, 0.2 s)

Measurement and execution

sparkMeasure StageMetrics

baseline_pipeline / compaction

Spark (local[*])

experiment_results.csv

1 create session (AQE off, broadcast -1, sparkMeasure JAR attached)

2 SparkSession

session warm-up

3 run every cell once, untimed, in random order

loop for each block b (A1, A2: 5 blocks; E6: 3 blocks)

4 shuffle cell order (seed 123 / 456 / 789)

loop for each cell (workload x setting or stage)

5 set spark.sql.shuffle.partitions (A2 stages: always 200)

6 run_and_measure(action_fn, metadata, warmup_runs=1, measured_runs=1)

loop 1 discarded warm-up, then 1 measured run

7 start()

8 begin()

9 action_fn() (timed with perf_counter)

10 read Parquet -> filter -> sort-merge join -> aggregate -> noop sink (or: coalesce + write compacted files)

11 done

12 return

13 end(); aggregate_stagemetrics()

14 executor run / CPU time, shuffle bytes, spill, GC, tasks

15 stop()

16 avg CPU %, peak memory %

17 2 rows (is_warmup flagged)

18 append_results(): measured row only, header checked against the 34-column schema

19 summarize_cells(): median, IQR, CV per cell (warn when CV > 0.20)

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experiment_results.csv